

MiMusicaa - PROJECT REPORT

Music Streaming Web Application

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Project Name: MiMusicaa

INTRODUCTION

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This is my project report for MiMusicaa, a music streaming web application similar to Spotify. This was my first full-stack project and I learned a lot!

WHAT I BUILT

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MiMusicaa is a website where users can:

- Listen to music online
- Search for songs
- Create an account and login
- See recently played songs
- Browse through 30 different songs

I made it look like Spotify with a dark theme and modern design.

TECHNOLOGIES USED

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Backend (Server):

- Node.js & Express.js - Server and API
- MongoDB - Database
- JWT & bcryptjs - Authentication and security
- Multer - File uploads

Frontend (User Interface):

- React - UI framework
- Vite - Development tool
- TailwindCSS - Styling
- Axios - API communication

MAIN FEATURES

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1. User Authentication

- Register with username, email, password
- Secure login with JWT tokens
- View profile information
- Logout functionality

2. Music Player

- Play/pause, next/previous controls
- Volume control
- Progress bar with seek
- Shows current time and duration
- Continues playing across pages

3. Music Library (30 Songs)

- Pop (7), R&B (7), Afrobeats (5), Hip-Hop (2)
- Electronic (3), Rock (1), Jazz (1), Alternative (1), Punjabi (1)
- Artists: The Weeknd (10 songs), SZA, Chris Brown, Wizkid, CKay, Rema, Burna Boy, Kendrick Lamar, Dua Lipa, Justin Bieber, Ed Sheeran

4. Recently Played

- Tracks play history
- Shows on home page
- User-specific

5. Search

- Search by title, artist, album, or genre
- Real-time results
- Grid display

6. User Interface

- Dark theme
- Responsive (mobile, tablet, desktop)
- Smooth animations
- Music cards with cover images

PROJECT STRUCTURE

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Backend (musicaa-api):

- controllers/ - Request handling logic
- models/ - Database schemas (User, Tracks, PlayHistory)
- routes/ - API endpoints (auth, tracks)
- uploads/ - Music files and cover images
- utils/seed.js - Database seeding script

Frontend (musicaa-webapp):

- components/ - MusicCard, Navbar, PlayerBar, Sidebar, Topbar
- pages/ - Home, Search

- contexts/ - PlayerContext for state management
- services/ - API calls

HOW TO RUN

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Backend Setup:

1. cd musicaa-api
2. npm install
3. Create .env file (MONGO_URI, JWT_SECRET, PORT=5001)
4. npm run seed
5. npm start

Frontend Setup:

1. cd musicaa-webapp
2. npm install
3. npm run dev
4. Open <http://localhost:5173>

CHALLENGES & SOLUTIONS

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- Audio Streaming: Implemented HTTP range requests for seeking
- Authentication: Used localStorage to persist login
- CORS Errors: Configured CORS properly
- File Uploads: Used Multer middleware
- State Management: Used React Context for player state

WHAT I LEARNED

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- Building REST APIs with Express
- MongoDB and database design
- JWT authentication and security
- React hooks and state management
- File handling and audio streaming
- Responsive design with TailwindCSS
- Debugging and problem-solving